

DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING
LINEAR INTEGRATED CIRCUITS AND APPLICATIONS
LAB

PROJECT REPORT

Multi-Functional Smart Monitoring and Control System

Submitted By

Name	Roll Number
B.V.Srinivasa Vaibhav	24ECB0A08
D.R.K.Ram Pawan Teja Reddy	24ECB0A15
K.Bheensen Reddy	24ECB0A29
S. Rupesh Sanjay	24ECB0A55

Submitted To

Dr. K. Vasundhara

Assistant Professor, Department of ECE

Academic Year: 2025–2026

1. Abstract

This report describes how we designed, simulated, and built a Multi-Functional Smart Monitoring and Control System. The system uses standard linear ICs and digital ICs to demonstrate several real-world sensing and control functions within a single integrated prototype.

The main circuit has three sub-modules: a temperature-controlled fan (NTC thermistor + LM358 op-amp comparator + 555 timer PWM driver); an IR-based direction detector (two IR sensors + two 555 monostable circuits + 7474 D flip-flop); and a buzzer-relay output stage (two 555 timers + BC547 transistor + flyback diodes).

Three additional standalone circuits are also included: an LDR-based automatic night light, a two-probe water level indicator with LED alerts, and a 00-to-99 event counter implemented using CD4033 decade counter ICs driving two seven-segment displays.

All circuits run from a single 9 V battery, regulated to 5 V through a 7805 IC. Component values were calculated from first principles, verified against datasheets, and confirmed on the physical hardware.

2. Components Used

2.1 Integrated Circuits

IC	Function	Qty
LM358 (Dual Op-Amp)	Comparator, buffer, constant-current source	3
NE555 Timer	PWM, monostable, tone generation	5
7474 (Dual D Flip-Flop)	Direction detection logic	1
7805 Voltage Regulator	9V to 5V regulated supply	1
CD4033 (Decade Counter + 7-Segment Driver)	IR event counter and direct display driving (00–99)	2

2.2 Passive Components

Component	Value / Rating	Qty
Resistor	10 k Ω	8
Resistor	1 k Ω	5
Resistor	100 k Ω	3
Resistor	330 Ω (LED current limiting)	8
NTC Thermistor	10 k Ω at 25 °C	1
LDR	$\approx$ 10 k Ω in dark	1
Electrolytic Capacitor	100 μ F / 25 V	4
Ceramic Capacitor	0.1 μ F (104)	4
Ceramic Capacitor	33 nF (333)	2

2.3 Active and Output Devices

Component	Specification	Qty
Diode – 1N4007	1 A, 1000 V – flyback protection	3
Transistor – BC547	NPN, 100 mA – driver stage	2
LED – Red	2 V, 20 mA – indication	3
LED – Green	2 V, 20 mA – indication	3
LED – White	3 V, 20 mA – night light	1
IR Sensor Module	Active-low output, 3–5 V supply	2
Active Buzzer	5 V DC, built-in oscillator	1
5 V Relay (SPDT)	5 V coil, 10 A contacts	1
5 V DC Fan	Brushless, ≈ 100 –200 mA	1
7-Segment Display (CC)	Common Cathode, red, 10-pin	2
Push Button Switch	Reset input	1

3. IC Specifications and Pin Details

3.1 LM358 – Dual Operational Amplifier

The LM358 has two independent op-amps in a single 8-pin DIP package. It works from a single supply of 3 V to 32 V, making it ideal for battery-powered designs. The output can swing close to ground, making it very useful in comparator mode.

Pin	Name	Function
1	OUT-A	Output of amplifier A
2	IN–A	Inverting input of amplifier A
3	IN+A	Non-inverting input of amplifier A
4	GND	Ground (0 V)
5	IN+B	Non-inverting input of amplifier B
6	IN–B	Inverting input of amplifier B
7	OUT-B	Output of amplifier B
8	VCC	Positive supply (3–32 V)

3.2 NE555 – Timer IC

The 555 is one of the most popular ICs ever made. In astable mode it produces a continuous square wave; in monostable mode it gives a single timed pulse. It runs from 4.5 V to 16 V and can source/sink up to 200 mA.

Pin	Name	Function
1	GND	Ground
2	TRIGGER	Starts timing – active LOW, $< V_{CC}/3$
3	OUTPUT	Output – HIGH or LOW
4	RESET	Active-LOW reset; tie HIGH if unused
5	CONTROL	Voltage control; bypass with $0.01\ \mu\text{F}$ to GND
6	THRESHOLD	Stops timing when $> 2/3 V_{CC}$
7	DISCHARGE	Discharges timing capacitor
8	VCC	Positive supply

3.3 7474 – Dual D Flip-Flop

The 74HC74 is a positive edge-triggered D flip-flop with active-low preset and clear inputs. We used it to latch which IR sensor was triggered first to determine the direction of movement.

Pin Group	Pins	Function
CLK1 / CLK2	3, 11	Rising-edge clock for each flip-flop
D1 / D2	2, 12	Data input
Q1 / Q2	5, 9	Normal output
$\overline{Q}1$ / $\overline{Q}2$	6, 8	Complemented output
PRE (active LOW)	1, 13	Asynchronous preset – tie HIGH if unused
CLR (active LOW)	4, 10	Asynchronous clear – tie HIGH if unused
VCC / GND	14, 7	Power supply

3.4 7805 – Voltage Regulator

The 7805 is a three-terminal fixed +5 V linear regulator with built-in current limiting and thermal shutdown. It supplies up to 1.5 A under proper heat dissipation. A 100 μ F electrolytic capacitor at the input and a 0.1 μ F ceramic capacitor at the output ensure stable and noise-free operation.

3.5 CD4033 – Decade Counter with 7-Segment Driver

The CD4033 is a Johnson decade counter with an integrated 7-segment decoder. It directly drives a common cathode seven-segment display, eliminating the need for a separate decoder IC.

Each clock pulse increments the count by one, and the IC internally converts the count into segment outputs (a–g) for display. A carry-out pin is provided to cascade multiple ICs for multi-digit counting.

Two CD4033 ICs are used in cascade to display counts from 00 to 99. The first IC drives the units digit, while its carry output is connected to the clock input of the second IC to drive the tens digit.

Pin	No.	Function
Clock	1	Pulse input
Reset	15	Resets count to zero
Carry Out	5	Drives next stage counter
Segment Outputs (a–g)	Various	Drive 7-segment display
VDD / VSS	16, 8	Power supply

4. Block Diagram Description

Note: Hand-drawn pin diagrams and block diagrams are included in the lab record. Signal flow is described below for clarity in the written report.

4.1 Power Supply Block

A 9 V battery is connected to the input of the 7805 voltage regulator. A 100 μF electrolytic capacitor is used for bulk filtering at the input, and a 0.1 μF ceramic capacitor is placed at the output to suppress high-frequency noise. The regulated 5 V supply is distributed to all ICs and output devices.

4.2 Temperature Monitoring Block (Main – Module 1)

LM358 #1 operates as a constant-current source for the NTC thermistor. LM358 #2 functions as a window comparator with two thresholds. When the temperature reaches the WARM level, the green LED turns ON. When the temperature crosses the OVERTEMP threshold, the red LED turns ON and releases 555 #1 from reset.

The 555 #1 timer operates in astable mode to generate a PWM signal, which drives the fan through a BC547 transistor. A 1N4007 diode is connected across the fan to protect against back-EMF.

4.3 IR Direction Detection Block (Main – Module 2)

Two IR sensing units are placed along the same path, spaced approximately 4 cm apart. Each detection event triggers a 555 timer configured in monostable mode, generating a clean pulse of approximately 1.1 s.

The output of 555 #2 is applied to the D input of the 7474 flip-flop, while the output of 555 #3 is connected to the clock input. Based on the order of triggering, the flip-flop stores the direction of motion. A HIGH output indicates left-to-right movement, while a

LOW output indicates right-to-left movement.

4.4 Buzzer and Relay Output Block (Main – Module 3)

555 #4 and 555 #5 operate in astable mode at different frequencies, producing two distinct tones for the active buzzer. The buzzer and relay are driven using BC547 transistors. Flyback diodes (1N4007) are connected across inductive loads to protect the transistors from voltage spikes.

4.5 IR Counter Block (Additional Circuit)

The IR counter is implemented using an IR LED transmitter and a receiver LED pair along with a BC547 transistor for signal conditioning. When the IR beam is interrupted, the transistor generates a clean pulse that is fed to the clock input of a CD4033 decade counter IC.

Two CD4033 ICs are cascaded to count from 00 to 99. The carry output of the first IC (units digit) is connected to the clock input of the second IC (tens digit). Each CD4033 directly drives a common cathode seven-segment display through current-limiting resistors. A push-button switch is used to reset the count to zero.

4.6 Additional Circuits Block

The **LDR Night Light** uses an LDR and a 10 k Ω resistor as a voltage divider. In darkness, the LDR resistance increases, causing the voltage at the divider node to rise above the threshold of the LM358 comparator. This switches the white LED ON. In bright light, the LDR resistance decreases, lowering the voltage and turning the LED OFF.

The **Water Level Indicator** uses two metallic probe wires placed at different heights. When water reaches a probe, it forms a conductive path due to dissolved ions in the water. This allows current to flow through a 10 k Ω resistor, activating the corresponding LED. The lower probe indicates minimum water level, while the upper probe indicates high water level.

The **IR Counter** is implemented by two CD4033 decade counter ICs. An IR LED and a receiver LED pair is used to detect beam interruption. When the beam is blocked, the BC547 transistor generates a pulse, which is fed to the clock input of first CD4033 (units digit). The carry output of the first CD4033 is connected to the clock input of the second CD4033 (tens digit),

Which enables counting from 00 to 99. Each CD4033 directly drives a common cathode seven-segment display through current-limiting resistors. A push-button switch is used to

reset both counters to 00.

5. Circuit Connection Tables

All connections are listed in the *From – To – Purpose* format and were used as our wiring checklist during assembly.

Table 5.1 Power Supply – 7805

From	To	Purpose
9V Battery (+)	7805 Pin 1 (INPUT)	Raw supply to regulator
9V Battery (–)	7805 Pin 2 (GND)	Common ground reference
9V Battery (+)	100 μ F Cap (+)	Input filter capacitor
7805 Pin 3 (OUTPUT)	0.1 μ F Cap (+)	Output bypass capacitor
7805 Pin 3	+5V Rail	Regulated 5V to all ICs

The 100 μ F electrolytic prevents input sag below the 7805 drop-out voltage. The 0.1 μ F ceramic keeps the 5 V rail clean for the 555 timers and CMOS logic.

Table 5.2 Thermistor Constant-Current Source – LM358 #1

From	To	Purpose
+5V via 100 k Ω	LM358 #1 Pin 3 (IN+A)	Set reference voltage
LM358 #1 Pin 1 (OUT-A)	NTC Thermistor Pin 1	Drive current into thermistor
NTC Thermistor Pin 2	LM358 #1 Pin 2 (IN–A)	Feedback for constant current
NTC Thermistor Pin 2	GND via 1 k Ω	Sense resistor sets current
NTC Thermistor Pin 1	Node-TEMP	Temp-proportional voltage

LM358 #1 forces $\approx 450 \mu$ A through the thermistor regardless of supply variations. Node-TEMP voltage reflects only temperature, not supply noise.

Table 5.3 Temperature Comparator and LED Indicators – LM358 #2

From	To	Purpose
Node-TEMP	LM358 #2 Pin 3 (IN+A)	Thermistor voltage input
10k+10k divider midpoint	LM358 #2 Pin 2 (IN−A)	WARM threshold (2.5 V)
LM358 #2 Pin 1 (OUT-A)	Green LED via 330 Ω	WARM indicator
100k+10k divider midpoint	LM358 #2 Pin 6 (IN−B)	OVERTEMP threshold (0.45 V)
LM358 #2 Pin 7 (OUT-B)	Red LED via 330 Ω	OVERTEMP indicator
LM358 #2 Pin 7	555 #1 Pin 4 (RESET)	Enable fan PWM only at OVERTEMP

Two comparators form a window detector. Comparator A catches the WARM condition; Comparator B detects OVERTEMP and simultaneously enables the fan PWM stage.

Table 5.4 PWM Fan Control – 555 #1 (Astable)

From	To	Purpose
555 #1 Pin 8	Pin 7 via 10 k Ω (Ra)	Timing resistor Ra
555 #1 Pin 7	Pin 6 via 1 k Ω (Rb)	Timing resistor Rb
555 #1 Pin 6	Pin 2 (TRIG)	Astable connection
555 #1 Pin 2	33 nF to GND	Timing capacitor
555 #1 Pin 3	BC547 Base via 1 k Ω	PWM to fan driver
BC547 Collector	Fan (−)	Fan current sink
1N4007 Anode	BC547 Collector	Back-EMF protection

555 #1 oscillates at ≈ 3.64 kHz with $\approx 91.7\%$ duty cycle, driving the fan at near-full speed when OVERTEMP is detected.

Table 5.5 IR Sensor Module Connections

From	To	Purpose
IR Module 1 – OUT	LM358 #3 Pin 3 (IN+A)	Buffer IR1 output
IR Module 2 – OUT	LM358 #3 Pin 5 (IN+B)	Buffer IR2 output
LM358 #3 Pin 1 (OUT-A)	555 #2 Pin 2 (TRIG)	IR1 triggers monostable 1
LM358 #3 Pin 7 (OUT-B)	555 #3 Pin 2 (TRIG)	IR2 triggers monostable 2

LM358 #3 buffers the open-collector IR module outputs to prevent the 555 trigger pins from loading them, which would cause unreliable triggering.

Table 5.6 Monostable Debounce and Direction Logic – 555 #2, #3, 7474

From	To	Purpose
+5V via 10 k Ω	555 #2 Pin 7	Timing resistor
555 #2 Pin 6	100 μ F to GND	Timing cap (≈ 1.1 s pulse)
555 #2 Pin 3 (OUT)	7474 Pin 2 (D1)	Clean pulse to D input
555 #3 Pin 3 (OUT)	7474 Pin 3 (CLK1)	Pulse clocks flip-flop
7474 Pin 5 (Q1)	Direction LED	HIGH = left-to-right
7474 Pin 6 ($\overline{Q}1$)	Reverse LED	LOW = right-to-left

The 1.1 s monostable window is wide enough for an object at walking speed to trigger both sensors before the pulse times out. The 7474 holds the direction result until the next object passes.

Table 5.7 Buzzer and Relay Output – 555 #4 and #5

From	To	Purpose
555 #4 Pin 3 (OUT)	BC547 Base via 330 Ω	Drive buzzer transistor
BC547 Collector	Buzzer (–)	Buzzer current sink
1N4007 across Buzzer	Collector to +5V	Spike protection
555 #5 Pin 3	Relay Coil via BC547	Relay driver
1N4007 across Relay Coil	Anode to Collector	Flyback suppression

The flyback diodes are essential — relay and buzzer coils generate large negative voltage spikes on de-energisation that would destroy the BC547 transistor without protection.

Table 5.8 LDR Night Light

From	To	Purpose
+5V	LDR Pin 1	Supply through LDR
LDR Pin 2	10 k Ω to GND	Voltage divider
Node-LDR	LM358 Non-Inverting Terminal	Darkness detection
LM358 OUT	White LED via 330 Ω	Night light output

In daylight, LDR resistance is $\approx 500\Omega$; at night it rises above 50 k Ω , crossing the comparator threshold and switching the LED on.

Table 5.9 Water Level Indicator

From	To	Purpose
+5V	Upper Probe	Reference voltage to water
Lower Probe	10 k Ω to GND	Pull-down and current limiting
Upper Probe Node	Red LED via 330 Ω	High water level indication
Lower Probe Node	Green LED via 330 Ω	Low water level indication

Water conducts electricity due to dissolved ions. When the water level reaches a probe, a conduction path is formed, allowing current to flow and turning on the corresponding LED. A $10\text{k}\Omega$ resistor limits the current to prevent corrosion of the probes and ensures safe operation.

Table 5.10 IR Counter 00–99

From	To	Purpose
IR LED (Transmitter)	Forward biased via $330\ \Omega$	Generates IR beam
IR Receiver LED	BC547 Base via $10\text{k}\Omega$	Detects beam interruption
BC547 Collector	+5 V via pull-up resistor	Signal conditioning
BC547 Output	CD4033 (Units) Clock Pin	Clean pulse for counting
CD4033 (Units) Carry Out	CD4033 (Tens) Clock Pin	Cascading for 2-digit count
CD4033 Outputs (a–g)	7-Segment Display via $330\ \Omega$	Display units digit
Second CD4033 Outputs	Second 7-Segment via $330\ \Omega$	Display tens digit
Reset Push Button	Reset pins of both CD4033 ICs	Reset count to 00

In this implementation, an IR LED transmitter and a receiver LED are used instead of a ready-made IR module. When the IR beam is interrupted, the signal at the BC547 transistor changes, generating a clock pulse for the CD4033 counter. The transistor acts as a signal conditioner to produce clean digital pulses. Two CD4033 ICs are cascaded to display counts from 00 to 99 on two seven-segment displays.

6. Working Explanation

6.1 Power Supply

The 9 V battery provides raw DC. The 7805 internally references 5 V and adjusts its pass transistor to keep the output steady. The 100 μF capacitor at the input absorbs sudden current demands and prevents the voltage from dropping below the drop-out threshold. The 0.1 μF ceramic on the output removes high-frequency transients that would otherwise cause erratic behaviour in the 555 timers and CMOS logic.

6.2 Temperature-Controlled Fan

- LM358 #1 forces a constant current ($\approx 450 \mu\text{A}$) through the NTC thermistor, keeping the reading independent of supply fluctuations.
- As temperature rises, NTC resistance drops, so Node-TEMP voltage falls.
- LM358 #2 Comparator A: when Node-TEMP crosses the WARM threshold ($\approx 2.5 \text{ V}$), the green LED turns on.
- LM358 #2 Comparator B: when Node-TEMP drops to the OVERTEMP threshold ($\approx 0.45 \text{ V}$), the red LED turns on and 555 #1's RESET is released.
- 555 #1 now oscillates at $\approx 3.64 \text{ kHz}$ with $\approx 91.7\%$ duty cycle, driving BC547 and the fan at near-full speed.
- The 1N4007 clamps back-EMF from the fan motor, protecting BC547.

6.3 IR Direction Detection

- Two IR sensing units are placed 4 cm apart along the same path.
- LM358 #3 buffers the outputs to ensure proper triggering.
- **Left to right:** first sensor triggers $\rightarrow$ 555 #2 fires $\rightarrow$ D of 7474 goes HIGH. Second sensor triggers $\rightarrow$ 555 #3 clocks 7474 $\rightarrow$ Q becomes HIGH.
- **Right to left:** second sensor triggers first $\rightarrow$ D remains LOW $\rightarrow$ 7474 clocks $\rightarrow$ Q becomes LOW.

6.4 Buzzer and Relay Output

555 #4 generates a tone that drives the active buzzer via BC547. 555 #5 drives the relay coil through another BC547 transistor. The relay can control external loads. Flyback diodes are used across both buzzer and relay to protect the transistors from voltage spikes.

6.5 LDR Night Light

The LDR and $10\text{ k}\Omega$ form a voltage divider. In daylight, LDR resistance is $\approx 500\ \Omega$ and the junction voltage is low. In darkness, LDR resistance increases above $50\text{ k}\Omega$, raising the junction voltage and switching on the white LED. The $0.1\ \mu\text{F}$ capacitor prevents flickering.

6.6 Water Level Indicator

Water acts as a weak conductor due to dissolved ions. When the water level reaches a probe, a conduction path is formed and current flows through the $10\text{ k}\Omega$ resistor, lighting the corresponding LED. The resistor also limits electrolysis current, reducing probe corrosion. The $330\ \Omega$ resistors limit LED current to approximately 9 mA .

6.7 IR Counter 00–99

An IR LED continuously emits light towards a receiver LED. When the beam is interrupted, the BC547 transistor switches and generates a pulse. This pulse is applied to the clock input of the first CD4033 IC (units digit).

After every 10 counts, the carry output of the first CD4033 is fed to the clock input of the second CD4033 (tens digit). Each CD4033 directly drives a common cathode seven-segment display through current-limiting resistors.

A push-button switch is connected to the reset pins of both ICs to manually reset the count to 00.

7. Mathematical Calculations

7.1 555 Astable Frequency – Fan PWM (555 #1)

$$f = \frac{1.44}{(R_a + 2R_b) \times C} \quad (1)$$

Given: $R_a = 10 \text{ k}\Omega$, $R_b = 1 \text{ k}\Omega$, $C = 33 \text{ nF}$

$$\begin{aligned} f &= \frac{1.44}{(10,000 + 2 \times 1,000) \times 33 \times 10^{-9}} \\ &= \frac{1.44}{12,000 \times 33 \times 10^{-9}} \\ &= \frac{1.44}{3.96 \times 10^{-4}} \\ &\approx \boxed{3,636 \text{ Hz} \approx 3.64 \text{ kHz}} \end{aligned}$$

Duty Cycle:

$$D = \frac{R_a + R_b}{R_a + 2R_b} \times 100 = \frac{11,000}{12,000} \times 100 \approx 91.7\% \quad (2)$$

The fan operates at near-full speed (91.7% ON time) when the OVERTEMP condition is reached.

7.2 555 Monostable Timing – Debounce (555 #2 and #3)

$$T = 1.1 \times R \times C \quad (3)$$

Given: $R = 10 \text{ k}\Omega$, $C = 100 \mu\text{F}$

$$T = 1.1 \times 10,000 \times 100 \times 10^{-6} = \boxed{1.1 \text{ seconds}} \quad (4)$$

This 1.1 s window is sufficient for an object moving at walking speed to trigger both IR sensors within the timing window.

7.3 LED Current-Limiting Resistor

$$R = \frac{V_{CC} - V_{LED}}{I_{LED}} \quad (5)$$

For Red/Green LEDs ($V_{LED} = 2\text{ V}$, $I_{LED} = 9\text{ mA}$):

$$R = \frac{5 - 2}{0.009} = \frac{3}{0.009} = 333\ \Omega \rightarrow \text{use standard } 330\ \Omega \quad (6)$$

Actual current:

$$I = \frac{5 - 2}{330} = 9.09\text{ mA} \quad \checkmark \quad (7)$$

For White LED ($V_{LED} \approx 3\text{ V}$):

$$R = \frac{5 - 3}{0.009} = 222\ \Omega \rightarrow \text{use } 330\ \Omega \text{ for safety} \quad (8)$$

Actual current:

$$I = \frac{5 - 3}{330} \approx 6.06\text{ mA} \quad \checkmark \quad (9)$$

7.4 Voltage Divider – Temperature Thresholds

WARM threshold ($10\text{ k}\Omega + 10\text{ k}\Omega$ divider):

$$V_{th} = V_{CC} \times \frac{R_2}{R_1 + R_2} = 5 \times \frac{10,000}{20,000} = 2.5\text{ V} \quad (10)$$

OVERTEMP threshold ($100\text{ k}\Omega + 10\text{ k}\Omega$ divider):

$$V_{th} = 5 \times \frac{10,000}{110,000} \approx 0.45\text{ V} \quad (11)$$

Since NTC voltage decreases as temperature rises, the OVERTEMP comparator detects when Node-TEMP falls *below* 0.45 V.

7.5 Thermistor Sense Current

$$I_{NTC} = \frac{V_{sense}}{R_{sense}} = \frac{0.45}{1,000} = 0.45\text{ mA} \approx 450\ \mu\text{A} \quad (12)$$

Self-heating power:

$$P = I^2 \times R = (0.45 \times 10^{-3})^2 \times 10,000 \approx 2\text{ mW} \quad (13)$$

This power is very small and does not significantly affect the thermistor reading.

7.6 IR Counter Segment Resistor

$$R_{seg} = \frac{V_{CC} - V_{LED}}{I} \approx \frac{5 - 2}{0.009} \approx 333 \Omega \rightarrow \text{use standard } 330 \Omega \quad (14)$$

External resistors ensure proper current limiting and uniform brightness of the seven-segment display.

8. Simulation and Analysis

Due to the mixed analog and digital nature of the project, different tools were used for analysis. Analog behaviour was studied using theoretical calculations, while selected modules were verified using simulation tools such as Tinkercad and Proteus.

Digital ICs such as CD4033 and 7474 are not directly supported in LTspice, hence full-system simulation was not performed. Instead, individual modules were analysed separately and validated through practical implementation.

8.1 Circuits Simulated and Analysed

- **LDR Night Light (Tinkercad):** Voltage divider and LED switching behaviour verified under light and dark conditions.
- **Water Level Indicator (Proteus):** Probe conduction and LED indication verified for different water levels.
- **Temperature Monitoring:** LM358 comparator analysed using theoretical calculations; NTC treated as variable resistor.
- **555 Astable (PWM):** $R_a = 10 \text{ k}\Omega$, $R_b = 1 \text{ k}\Omega$, $C = 33 \text{ nF}$ – frequency and duty cycle calculated.
- **555 Monostable (Debounce):** $R = 10 \text{ k}\Omega$, $C = 100 \mu\text{F}$ – pulse width calculated.

8.2 Expected Results

Circuit	Parameter	Expected Output
555 Astable	Output Pin 3	Square wave, ≈ 3.64 kHz, 91.7% duty
555 Monostable	Output Pin 3	Single HIGH pulse, ≈ 1.1 s width
LM358 Comparator	Output	Switching at threshold voltage
LDR Divider	Node voltage	Increases in dark, decreases in light
BC547 Switching	Collector voltage	Switches between 0 V and V_{CC}

The calculated and simulated behaviours were compared with practical observations. Minor deviations were observed due to component tolerances and manual wiring, but overall system behaviour matched expected results.

9. Theoretical vs. Practical Readings

Parameter	Theoretical	Observed (Practical)	Remarks
555 #1 Frequency	3.64 kHz	≈ 3.5 – 3.7 kHz	Minor variation due to capacitor tolerance
Monostable Pulse Width	1.1 s	≈ 1 s	Slight variation due to RC tolerance
WARM Threshold	2.5 V	≈ 2.4 – 2.6 V	Comparator offset and resistor tolerance
OVERTEMP Threshold	0.45 V	≈ 0.4 – 0.5 V	Sensitive to resistor ratio
LED Current ($330\ \Omega$)	9.09 mA	≈ 8 – 9 mA	Within safe operating range
7805 Output Voltage	5.0 V	≈ 4.9 – 5.0 V	Stable regulated output
Fan Operation	ON at high temperature	Works with slight delay	Due to thermal response time
Counter Operation	Counts 00–99	Mostly works with occasional errors	Noise and wiring issues

The observed values are close to theoretical calculations. Small deviations are mainly due to component tolerances (resistors $\pm 5\%$, capacitors $\pm 20\%$), wiring effects, and environmental

conditions. The counter section showed minor instability due to noise and loose connections in the prototype setup.

10. Applications

Sub-Module	Real-World Applications
Temperature-Controlled Fan	Server room cooling, transformer cooling, incubator ventilation
IR Direction Detection	Automatic door control, people counting, conveyor belt sorting
Buzzer and Relay Output	Alarm systems, home automation switching, machine interlocks
LDR Night Light	Street lighting automation, staircase lighting, bedside lamps
Water Level Indicator	Tank monitoring, sump pump control, aquarium level indication
IR Counter (00–99)	Visitor counting, object counting in production lines

11. Design Methodology

We followed a structured bottom-up approach throughout the project:

1. **Requirement Analysis:** All required sensing and control functions were identified, and suitable analog and digital ICs were selected for each module.
2. **Component Selection:** Components were chosen based on availability and compatibility with a common 5 V supply using the 7805 regulator.
3. **Manual Calculations:** Timing parameters, LED current-limiting resistors, voltage divider thresholds, and thermistor current were calculated before implementation.
4. **Simulation and Analysis:** Selected modules were analysed using simulation tools. The LDR circuit was verified in Tinkercad, and the water level indicator was tested in Proteus. Other circuits were analysed theoretically due to mixed analog and digital components.
5. **Breadboard Prototyping:** Individual sub-modules were first tested separately to verify basic functionality before integration.

6. **Hardware Implementation:** The circuits were assembled on a general-purpose PCB. Practical challenges such as wiring complexity and noise were observed during this stage.
7. **Debugging and Adjustment:** Adjustments were made to improve stability, including checking connections, improving grounding, and adjusting sensor placement.
8. **Verification:** Theoretical values were compared with observed behaviour, and deviations were analysed based on component tolerances and practical limitations.

12. Problems Faced

12.1 Loose Connections on PCB

During testing, some modules behaved inconsistently. The fan and counter sections did not respond properly at times. On inspection, we found that loose solder joints and wiring issues were the main cause. Re-soldering and tightening connections improved stability.

12.2 IR Detection Instability

Instead of IR modules, we used an IR LED and receiver LED with a BC547 transistor. Due to ambient light and alignment issues, the signal was not always clean, leading to missed or false counts. Proper alignment and shielding reduced the problem but did not eliminate it completely.

12.3 Monostable Timing Variation

The $100\mu\text{F}$ capacitor used in the 555 monostable circuit showed variation in actual value, resulting in slight deviation from the expected pulse width of 1.1 s. This caused inconsistent triggering in some cases.

12.4 7805 Heating Issue

When multiple modules were active, the total current increased, causing the 7805 regulator to heat up. A small heatsink was added to improve heat dissipation and maintain stable operation.

12.5 Seven-Segment Display Issues

The seven-segment display driven by CD4033 showed inconsistent brightness in some segments. This was mainly due to unequal current distribution and wiring complexity. Adding proper current-limiting resistors improved the display visibility.

12.6 Sensor Calibration Difficulty

The NTC thermistor response varied slightly from expected values, making it difficult to fix an exact temperature threshold. Manual adjustment of the comparator reference was required to achieve approximate switching.

12.7 Noise and Wiring Complexity

Since the circuit was implemented on a general-purpose PCB with many interconnections, noise and wiring complexity affected the overall performance, especially in the counter section. Proper grounding and shorter connections helped reduce this issue.

13. Limitations

- The temperature control threshold is fixed by resistor values. Any change in trigger temperature requires manual adjustment or replacement of components.
- The IR counter does not have memory storage. The count resets to zero when power is turned off.
- The IR detection system is sensitive to alignment and ambient light, which can cause missed counts or false triggering.
- The system performance is affected by noise and wiring complexity, especially in the counter section, due to the use of a general-purpose PCB.
- The 7805 voltage regulator may heat up when multiple modules operate simultaneously, limiting the maximum load current.
- The LDR-based night light depends on surrounding light conditions and may require adjustment for different environments.
- The prototype is built on a PCB/cardboard base and is not suitable for long-term or outdoor use without proper enclosure and protection.

14. Cost Estimation

Component	Qty	Unit (Rs.)	Total (Rs.)
LM358 (DIP-8)	3	12	36
NE555 Timer	5	8	40
7474 Dual D Flip-Flop	1	15	15
CD4033 Counter IC	2	25	50
7805 Voltage Regulator	1	10	10
NTC Thermistor 10 k Ω	1	15	15
LDR	1	5	5
IR LED + Receiver LED	2 pairs	10	20
BC547 Transistor	2	3	6
1N4007 Diode	3	2	6
Resistors (assorted)	1 set	40	40
Capacitors (assorted)	1 set	30	30
LEDs (Red, Green, White)	7	3	21
Active Buzzer (5 V)	1	15	15
5V Relay	1	25	25
5V DC Fan	1	60	60
7-Segment Display (CC)	2	20	40
9V Battery + Clip	1	30	30
PCB / Breadboard	1	100	100
Wires, Connectors, Misc.	1 set	200	200
Total Estimated Cost			759

The estimated cost based on individual components is approximately **Rs. 750–800**. However, the actual expenditure for the complete project was around **Rs. 900**, considering additional miscellaneous expenses such as extra components, replacements, and wiring materials.

15. Conclusion

This project demonstrated a multi-functional smart monitoring and control system using standard linear and digital ICs. The system integrates multiple modules including temperature-based fan control, IR direction detection, buzzer and relay output, LDR-based night light, water level indication, and a 00–99 IR counter.

All modules were implemented on a cardboard-based prototype and tested individually. While most modules functioned as expected, some sections such as the IR counter and detection circuits showed minor instability due to noise, alignment issues, and wiring complexity.

The deviations between theoretical and observed values were within acceptable limits, mainly influenced by component tolerances and practical implementation factors. During the project, several challenges such as unstable connections, sensor alignment, and regulator heating were encountered and partially resolved, which improved our debugging and practical understanding.

The total project cost was approximately **Rs. 900**, making it a low-cost solution for demonstrating multiple sensing and control applications. With further improvements such as proper PCB design, better signal conditioning, and enclosure, the system can be made more reliable for real-world applications.

Overall, this project strengthened our understanding of linear ICs, digital logic, and their practical implementation, fulfilling the core objectives of the LICA laboratory course.